

MACHINE LEARNING • BAB 5

Decision Tree dan Pruning

5.11 Implementasi Python: Praktikum 5.1 sampai 5.6

Ahmad Dandi Subhani

202343500126 • Kelas R7B

PENDAHULUAN

Tujuan Praktikum

- Memahami cara kerja **decision tree** pada data nyata
- Membandingkan tiga skenario: **tanpa pruning**, **pre-pruning**, dan **post-pruning (CCP)**
- Menjawab: strategi pruning mana yang terbaik, dan kenapa?

DATASET

Breast Cancer Wisconsin

- **569 pasien**, 30 fitur numerik
- 2 kelas: **malignant** (212) dan **benign** (357)
- Sumber: **sklearn.datasets**, tanpa unduhan eksternal

Ukuran data: (569, 30)

Distribusi kelas: [212 357]

PRAKTIKUM 5.1

Persiapan Library

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, validation_curve
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report
```

- Tiga kelompok: **komputasi dan visualisasi**, **data**, **model dan evaluasi**

PRAKTIKUM 5.2

Import Dataset

```
data = load_breast_cancer(as_frame=True)
X, y = data.data, data.target
print("Ukuran data:", X.shape)
print("Distribusi kelas:", np.bincount(y))
```

```
Ukuran data: (569, 30)
Distribusi kelas: [212 357]
```

- Dataset Breast Cancer Wisconsin langsung dari sklearn, **tanpa unduh file**

PRAKTIKUM 5.3

Preprocessing: Split Data

```
X_train, X_test, y_train, y_test = train_test_split(X, y,  
                                                    test_size=0.2, random_state=42, stratify=y)
```

- 80% latih, 20% uji
- **stratify=y**: proporsi kelas tetap sama di kedua sisi
- **random_state=42**: hasil split selalu sama, eksperimen bisa diulang

Training Tiga Model

```
# Model tanpa pruning (tumbuh penuh)
tree_full = DecisionTreeClassifier(random_state=42)
tree_full.fit(X_train, y_train)

# Model dengan pre-pruning
tree_pruned = DecisionTreeClassifier(max_depth=4,
    min_samples_leaf=5, random_state=42)
tree_pruned.fit(X_train, y_train)

# Model dengan post-pruning (Cost Complexity Pruning)
path = tree_full.cost_complexity_pruning_path(X_train, y_train)
ccp_alpha_terpilih = path.ccp_alphas[len(path.ccp_alphas)//2]
print("ccp_alpha terpilih:", ccp_alpha_terpilih)
tree_ccp = DecisionTreeClassifier(ccp_alpha=ccp_alpha_terpilih,
    random_state=42)
tree_ccp.fit(X_train, y_train)
```

```
ccp_alpha terpilih: 0.005274725274725275
```

PRAKTIKUM 5.5

Evaluasi

```
for nama, model in [("Tanpa Pruning", tree_full),  
    ("Pre-Pruning", tree_pruned), ("Post-Pruning (CCP)", tree_ccp)]:  
    acc_train = accuracy_score(y_train, model.predict(X_train))  
    acc_test = accuracy_score(y_test, model.predict(X_test))  
    print(f"{nama}: Akurasi Train={acc_train:.3f}, "  
        f"Akurasi Test={acc_test:.3f}, Gap={acc_train-acc_test:.3f}")
```

```
Tanpa Pruning: Akurasi Train=1.000, Akurasi Test=0.912, Gap=0.088  
Pre-Pruning: Akurasi Train=0.976, Akurasi Test=0.921, Gap=0.055  
Post-Pruning (CCP): Akurasi Train=0.978, Akurasi Test=0.930, Gap=0.048
```

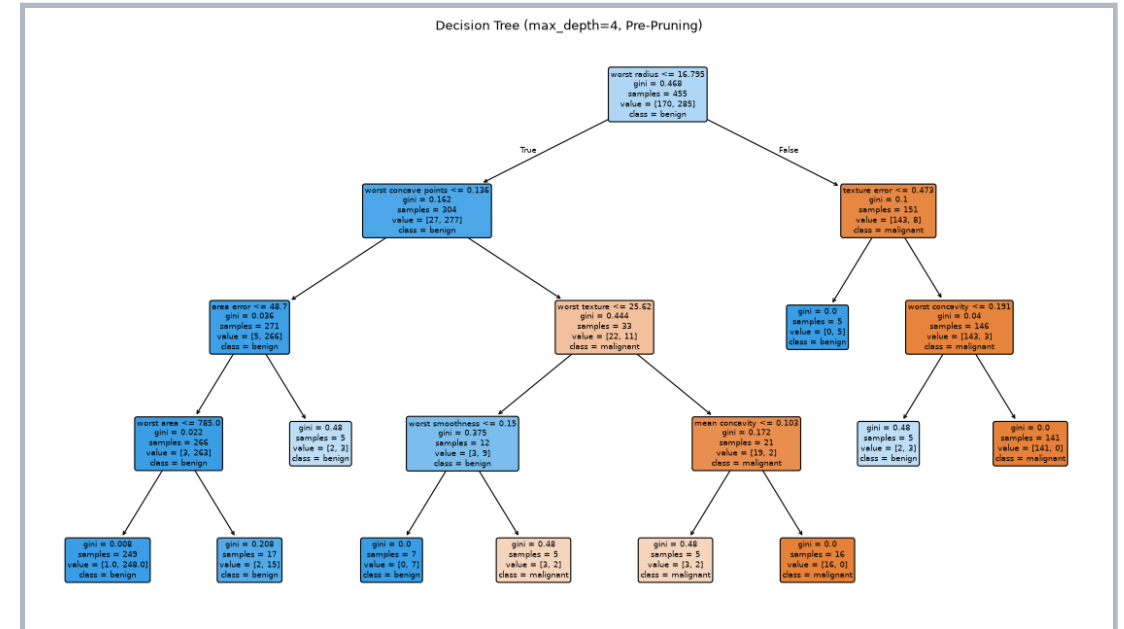
Gap = train minus test. Gap besar artinya **overfitting**. CCP: test tertinggi, gap terkecil.

PRAKTIKUM 5.6

Visualisasi Pohon (Pre-Pruning)

```
plt.figure(figsize=(14,8))
plot_tree(tree_pruned,
          feature_names=data.feature_names,
          class_names=data.target_names,
          filled=True, rounded=True,
          fontsize=7)
plt.title("Decision Tree
(max_depth=4, Pre-Pruning)")
plt.show()
```

- Biru = **benign**, oranye = **malignant**
- Tiap jalur terbaca sebagai aturan **if-then**



5.11.1 INTERPRETASI

Apa Artinya?

- Tanpa pruning: train **1,000**, gap **0,088**. Ini pola khas **overfitting**
- Pruning menukar sedikit akurasi train dengan generalisasi yang jauh lebih baik
- CCP terbaik: test **0,930**, gap **0,048**, hanya 7 daun
- Pemilihan parameter idealnya lewat **validation curve** atau **cross-validation**

Latihan 1: Grid Search Pre-Pruning

- **12 kombinasi** diuji: max_depth
{2, 4, 6, None} x min_samples_leaf
{1, 5, 10}
- Akurasi test tertinggi **0,9474**,
dicapai tiga kombinasi: (4, 10), (6,
10), (None, 10)
- Pola konsisten:
min_samples_leaf=10 selalu
menang di tiap max_depth

5.12 LATIHAN PRAKTIKUM

Latihan 2: Cost Complexity Pruning Path

- **Alpha optimal = 0,002866:**
akurasi test 0,9386 dengan 12 daun
- Sebelum pruning: 19 daun (0,9123). Sesudah: **12 daun (0,9386)**, lebih kecil dan lebih akurat
- Kurva naik lalu turun: pruning berlebih (1 daun) anjlok ke **0,6316**, underfitting

KESIMPULAN

Tiga Poin Kunci

- **Pruning mengurangi overfitting** sekaligus menyederhanakan model
- **Model sederhana bisa menang**: 7 daun mengalahkan 19 daun di data uji
- **Decision tree unggul di interpretabilitas** dibanding model black-box

BAB 5 • DECISION TREE DAN PRUNING

Terima Kasih

Ahmad Dandi Subhani

202343500126 • Kelas R7B